

Role 1: Data Ingestion Specialist

Project:	MOSPI Airfare Index	Route Corridor:	DEL-BOM (T+1 to T+5)
Tech Stack:	Python 3.13, Playwright, JSONL	Status:	VERIFIED & LOCKED (Sept 3, 2026)

1. High-Level Architecture

Role 1 is the edge ingestion engine designed to acquire domestic airfares at scale. Instead of using fragile HTML DOM parsing, it utilizes Playwright Chromium to intercept background XHR/Fetch API JSON responses directly from network traffic.

2. System Component Summary

Component / File	Core Mechanism	Key Output / Benefit
Matrix Generator	Relative offset arithmetic	Dynamic targets for T+1 to T+5 departure dates
Stealth Browser	AutomationControlled flag removal	Bypasses Cloudflare / Akamai bot rejections
Network Interceptor	Async listener (page.on('response'))	Direct raw API JSON payload capture
Staging Persistence	Append-only JSONL streaming	Audit-ready contract for Role 2 Data Engineering

3. Staging Schema Contract (seed_data/staging_raw_payloads.jsonl)

{ "scraped_at": "2026-08-27T15:45:00.000Z", "route_code": "DEL-BOM", "advance_window": "T+1", "departure_date": "2026-08-28", "raw_payload": { "carrier": "IndiGo", "flight_no": "6E-201", "total_quote": 6420.0, "fare_details_html": "Base: 5186, Tax: 933.48" } }

4. Verification Protocol & Audit Metrics

- * Execution Command: python src/ingestion/scrapper_lead.py
- * Interception Success Rate: 100% (5 out of 5 network API calls captured)
- * Bot Block / Anti-Scraping Rate: 0% (Zero session rejections recorded)
- * Data Output Contract: 5 valid JSONL records written to staging directory